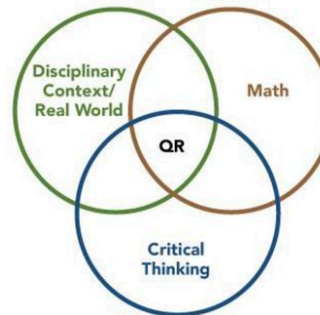


CS 160 – Exploring Computer Science

Assignment 4: Quantitative Reasoning (Data)

What is Quantitative Reasoning?

- Also known as quantitative literacy
- The application of mathematical thought & knowledge to authentic, everyday issues
- Demonstrated by the ability to make reasoned decisions using mathematical knowledge



Purpose

In this assignment you will be provided a dataset, and asked to obtain insights and information, really anything of value out of data. You will also draw insightful conclusions and consider the societal impact of the information provided by your charts.

In order to do this, we start with a question that you can answer with the dataset, identify the columns of information needed to answer the question, represent the data using graphs or charts, and interpret the charts and come to conclusions.

After completing this assignment, you will be able to:

- Draw conclusions from data by using quantitative reasoning, including reading charts, graphs, or tables.
- Explain the societal impact of the Data Analysis using insights and information obtained from the dataset.

Task

This assignment is asking you to find a dataset and perform some analysis about it. This is meant to be an act of creative exploration: you can pick a topic that is serious or silly. The site I'm suggesting for finding a dataset has everything from cups of coffee consumed in the 1990s show Twin Peaks, to coral reef health in the pacific, to data that was used to track down the fate of migrant children who were placed in the homes of strangers.

I highly recommend finding a dataset that has a .csv file you can download as this is the easiest form of data to analyze

Follow the steps below and fill out the questions when prompted:

1. State the Problem or Question

Find a dataset via the archives of the excellent newsletter series Data Is Plural:
<https://www.data-is-plural.com/archive/> and copy the URL of the dataset you use below
What dataset did you choose? Please state an interesting question you would like to answer using this dataset and predict what the answer will be.

2. Determine Information Needed

Please answer questions a-d below:

- a. What information will you need to answer your question? Which columns of data will be relevant?
- b. Explain why this information is important.
- c. Is the data quantitative (is it discrete or continuous) or categorical?
- d. What type of charts (bar chart, donut chart, histogram, scatter plot, or crosstab) will you use to visualize the data?

3. Representation

Please open the dataset in Google Colab Notebook (or other application of your choosing) and use the charting tools. Add appropriate labels for the x-axis, y-axis, legend (if appropriate), and graph title. Please copy / paste the charts below that you described in question 2d.

4. Interpretation

What patterns in your charts from step 3 address your question from step 1?

In your response, make sure to:

- a. Use mathematical terms (percents, counts, averages, ranges, etc.)
- b. Consider correct use of units of measurement (percents, counts, money, prediction scores, etc.)
- c. Identify any assumptions you've made in your interpretation

5. Conclusions

Does your interpretation in step 4 actually answer your question from step 1? Here are some limitations to consider:

- a. Does your interpretation of the data align with your prediction from step 1? Why or why not?
- b. Often we cannot be sure whether a pattern we observe in a chart (a correlation) means that one of the columns is causing the other. Is that the case here, why or why not?
- c. Discuss any inconclusive or limited results. Would modifying your question lead to a more conclusive result?

6. Social Impact

What communities could be affected by the conclusions you've drawn? Explain how they might react to the analysis you've done here.